SUR4376 / SUR6377 Geospatial Applications of UAS

Fall, 2025
Hybrid Synchronous, 3 Credits

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Office Hours: Wednesday (3:00–5:00PM) Thursdays (1:30–2:30PM) in-person or via web conferencing. Please schedule an appointment by emailing the instructor to ensure availability.

Class Meeting Times
Wed. 7th period (1:55-2:45), Fri. 6-7 period (12:50-2:45)

Course Description
Covers contemporary issues and common applications associated with small UASs (Unmanned Aerial Systems).

Course Learning Objectives
This covers issues and geospatial applications associated with small unmanned aerial systems (UASs). By the end of this course, the student will be able to:

- describe common/typical UAS applications (e.g. agriculture, forestry, inspection)
- evaluate the technical considerations involved in implementing UAS applications
- cite the common business and legal aspects of operating UASs
- lead and participate in current debates on UAS issues and applications

Course Overview
This course centers around a series of discussion sessions and expert guest lectures. Topics and speakers in the course represent a wide swath of potential applications of the maps and other geospatial products that can be captured from small UAS.

Course Prerequisites
None.

Learning Materials
Learning material is provided the week before each topic, some provided by the guest speaker, some by the instructor. Material comprises journal articles, videos, websites and other media, and will be posted to the weekly modules for student access.

Interaction Plan
The class meets twice a week: a one period synchronous online session on Wednesday (1:55pm-2:45pm) for a preparatory discussion; and a two-period in-person (GNV sections) and online synchronous session on Friday (12:50pm-2:45pm) for the lecture. This class is designed as an active learning experience. In-class and online discussions are critical, and students are expected to engage in these activities.

Students prepare for each weekly guest lecture by completing related study assignments and doing independent research. In addition, each week a facilitator or facilitator-group, is tasked to lead a discussion in preparation for the guest lecture. This serves to increase the class' foundational knowledge in the area to be presented and facilitates active discourse during the guest lecture. A different person or team will guide the preparatory discussion each week.

Students are expected to participate via the message boards on the course website, which serves as the pool of topics and questions to guide the Q/A portion of the speakers' presentations.

Students also, with guidance from the instructors, develop a term project (see below) to be presented towards the end of the semester.

Required Technology

- A computer or mobile device with high-speed internet connection.
- A webcam, headset and/or microphone, and speakers.
- Latest version of web browser. Canvas supports only the two most recent versions of any given browser. What browser am I using?

Communication Guidelines

Out-of-class online communication with the instructor should be through the course website via Canvas email (Inbox).

Class Demeanor/Expectations

Just as in any professional environment, meaningful and constructive dialogue is expected in this class and requires a degree of mutual respect, willingness to listen, and tolerance of opposing points of view. Respect for individual differences and alternative viewpoints will be maintained in this class at all times. All members of the class are expected to follow rules of common courtesy, decency, and civility in all interactions. Failure to do so will not be tolerated and may result in loss of participation points and/or referral to the Dean of Students' Office.

Technical Support

UF Computing Help Desk & Ticket Number: All technical issues require a UF Helpdesk Ticket Number. The UF Helpdesk is available 24 hours a day, 7 days a week. https://helpdesk.ufl.edu/ | 352-392-4357

2025 Modules

#	SPEAKER	TOPIC(S)	Organization	Wed. Prep Session	Friday Lecture
0	Dr. Ben Wilkinson	Intro and UAS Mapping Basics	UF, Geomatics	none	22-Aug
1	John Rouse	UAS Rules and Regulations	UF, EH&S	27-Aug	29-Aug
2	Bret Bienkowski	Commercial Applications	Tetra Tech	3-Sep	5-Sep
3	Allix North	Ecology, Coastal Management	FL DEP	10-Sep	12-Sep
4	Dr. Peter Ifju	Vehicle Design	UF, Aerospace Eng.	17-Sep	19-Sep
5	Dr. Matthew Burgess	USGS Applications	USGS	24-Sep	26-Sep
6	Dr. Eben Broadbent	Tropical Forestry	UF, SFFGS	1-Oct	3-Oct
7	Dr. Josh Rudd	Agricultural Applications	Syngenta	8-Oct	10-Oct
Homecoming – Wednesday is Optional Term Project Discussion No Speaker Friday					17-Oct
8	Casey Shanks	NGA Applications	NGA Geodetic Surveys	22-Oct	24-Oct

9	Dr. Mike Starek	Transportation and AI	TAMUCC, Mantis	29-Oct	31-Oct

Weekly Course Schedule

Week	Topic	Assessment	Due Dates
1	Module 0	Discussion Board Introductions Lect. Date Preference Poll	Aug. 25 Aug. 25
2	Module 1	Discussion Board Post	Aug. 27
3	Module 2	Discussion Board Post Term Project Proposal	Sep. 3 Sep. 5
4	Module 3	Discussion Board Post	Sep. 10
5	Module 4	Discussion Board Post	Sep. 17
6	Module 5	Discussion Board Post Quiz 1	Sep. 24 Sep. 24
7	Module 6	Discussion Board Post	Oct. 1
8	Module 7	Discussion Board Post	Oct. 8
9	Term Project Discussion Wed.	None.	Oct. 15 (optional)
10	Module 8	Discussion Board Post	Oct. 22
11	Module 9	Discussion Board Post	Oct. 29
12	No Class	Quiz 2	Nov. 5
13	Term Project Presentations (Wed.)	PPT Slide Decks Due Presentations	Nov. 12 Nov. 12 (varies)
14	Term Project Presentations (Wed., Fri.)	Presentations Presentations	Nov. 19 (varies) Nov. 21 (varies)
15		Term Project Papers Due (grads)	Dec. 3

Grading Policy

Course grading is consistent with UF grading policies.

Late Policy

Due dates will be announced when assignments are given. A deduction of 10% will be made for each day they are late. It is the responsibility of the student to access on-line lectures, readings, quizzes, and exams and to maintain satisfactory progress in the course. Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at: https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx

Course Grading Structure

The course grade will be based on participation, labs, reading assignment quizzes, computer programming homework (SUR 6355) and three exams. The percentage breakdown is as follows (subject to change based on course progression). Final grades are based on a curve.

Assignment Type	Description	Percent of Final Grade
Participation	Attendance, Discussion Sessions, Boards	25%
Prep Presentation	Wednesday Session Lead and Q/A Facilitation, Lecture Notes Posted **READ THE ASSIGNMENT DESCRIPTION IN CANVAS CAREFULLY!**	30%
Quizzes	Two open-note quizzes on the guest speaker topics	20%

Assignment Type	Description	Percent of Final Grade
Term Project	Presentation of individual student projects, grad students also submit a paper	25%

Grading Scale

Grade	Percentage
A	94-100
A-	90-93.99
B+	87-89.99
B	83-86.99
B-	80-82.99
C+	77-79.99
C	73-76.99
C-	70-72.99
D+	67-69.99
D	63-66.99
D-	60-62.99
E	0-59.99

Policy on AI Use

Although it can be useful as a tool for some things, generative AI can be unreliable and doesn't always provide correct information. On certain assignments, the instructor will explicitly either allow or disallow the use of generative AI, such as ChatGPT, in completing the work. If it is explicitly allowed, students must thoroughly cite any information or assistance received from AI/NLP platforms, just as you must for any resource. If the instructor has not mentioned the use of this technology for a given assignment, the student should assume it is entirely prohibited. When in doubt, students should ask the instructor if and how AI/NLP may be used to avoid the risk of losing up to all points on the associated assignment. This policy is not a substitute for the policy of any other instructor or course. All courses will have their own individual policy, and students should learn and understand the policy for any course they are taking.

Academic Policies and Resources

Academic policies for this course are consistent with university policies. See https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

Campus Health and Wellness Resources

Visit https://one.uf.edu/whole-gator/topics for resources that are designed to help you thrive physically, mentally, and emotionally at UF.

Please contact UMatterWeCare for additional and immediate support.

Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Privacy and Accessibility Policies

[required for online courses, list all technology used]

- Instructure (Canvas)
  - Instructure Privacy Policy
  - Instructure Accessibility
- Zoom
  - Zoom Privacy Policy
  - Zoom Accessibility